

CSE 150A-250A AI: Probabilistic Models

Lecture 12

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Slides adapted from previous versions of the course (Prof. Lawrence, Prof. Alvarado, Prof Berg-Kirkpatrick)

Agenda

Review

Viterbi algorithm

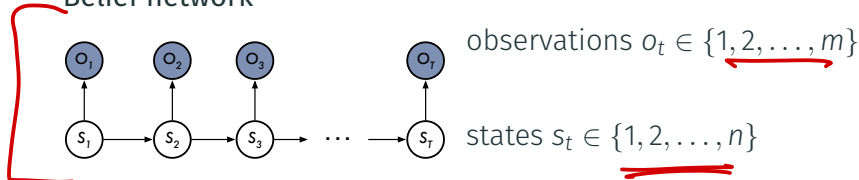
Learning in HMMs

Backward Algorithm

Review

Hidden Markov models

- Belief network



- Parameters

$$a_{ij} = P(S_{t+1}=j|S_t=i)$$

transition matrix

$$b_{ik} = P(O_t=k|S_t=i)$$

emission matrix

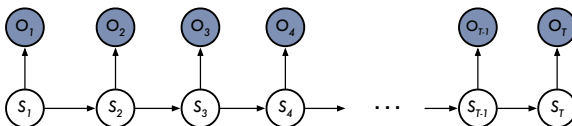
$$\pi_i = P(S_1=i)$$

initial state distribution

- Notation

Sometimes we'll write $b_i(k) = b_{ik}$ to avoid double subscripts.

Key computations in HMMs



Inference

1. How to compute the likelihood $P(o_1, o_2, \dots, o_T)$?
2. How to compute the most likely hidden states $\operatorname{argmax}_{\vec{s}} P(\vec{s} | \vec{o})$?
3. How to update beliefs by computing $P(s_t | o_1, o_2, \dots, o_t)$?

Learning

How to estimate parameters $\{\pi_i, a_{ij}, b_{ik}\}$ that maximize the log-likelihood of observed sequences?

forward

Decoding

Forward Algorithm

For a particular sequence of observations $\{o_1, o_2, \dots, o_T\}$,
define the matrix with elements:

define the matrix with elements:

$$\alpha_{it} = P(o_1, o_2, \dots, o_t, S_t=i)$$

Handwritten notes: t (above the matrix), i (to the left of the matrix), n rows (to the left of the matrix), and a red double underline under α_{it} .

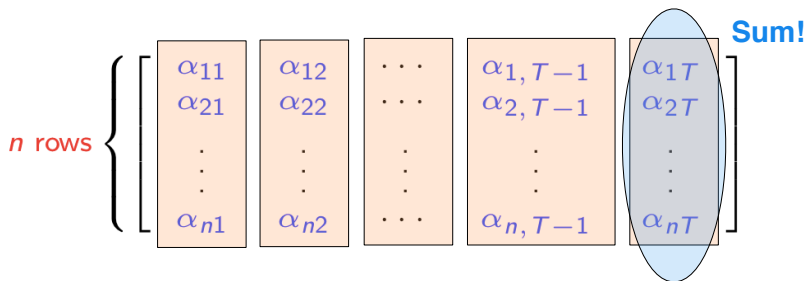
α_{11}	α_{12}	$\dots$	$\alpha_{1,T-1}$	α_{1T}
α_{21}	α_{22}	$\dots$	$\alpha_{2,T-1}$	α_{2T}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
α_{n1}	α_{n2}	$\dots$	$\alpha_{n,T-1}$	α_{nT}

The forward algorithm fills in the matrix of α_{it} elements one column at a time:

$$\alpha_{i1} = \pi_i b_i(o_1)$$

$$\alpha_{j,t+1} = \sum_{i=1}^n \alpha_{it} \underbrace{a_{ij}} \underbrace{b_j(o_{t+1})}$$

Computing the likelihood $P(o_1, o_2, \dots, o_T)$



$$P(o_1, o_2, \dots, o_T)$$

$$= \sum_{i=1}^n P(o_1, o_2, \dots, o_T, S_T = i)$$

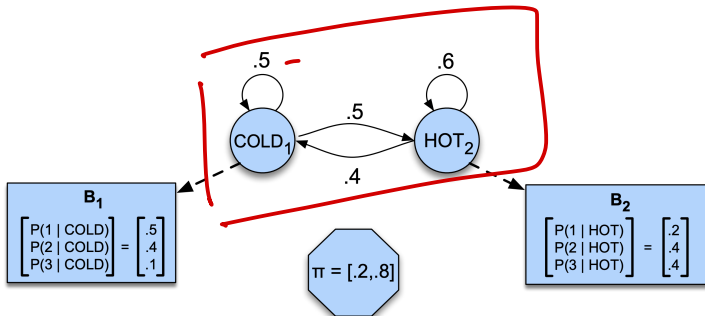
marginalization

$$= \sum_{i=1}^n \alpha_{iT}$$

sum of last column

Example¹

- Two hidden states (Weather): $\{H, C\}$
- Observations (Ice creams): $\{1, 2, 3\}$

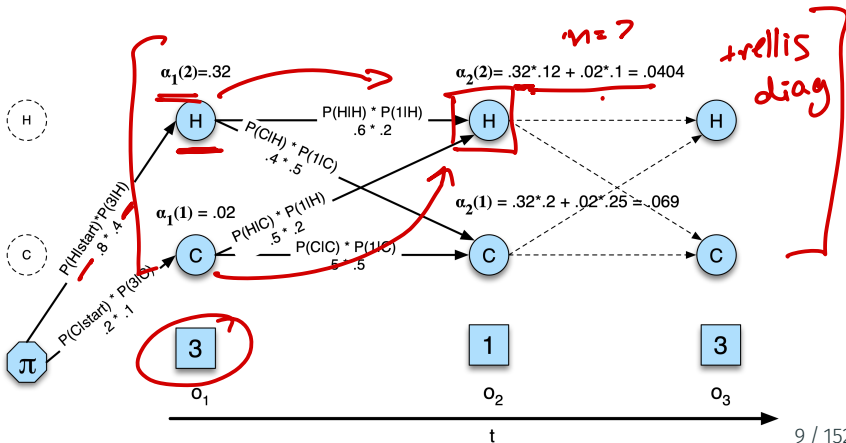


¹Eisner, J. 2002. An interactive spreadsheet for teaching the forward-backward algorithm.

Example - Forward Algorithm

$$\alpha_{i1} = \pi_i b_i(o_1)$$

$$\alpha_{j,t+1} = \sum_{i=1}^n \alpha_{it} a_{ij} b_j(o_{t+1})$$



Viterbi algorithm

The most likely sequence of hidden states

$$\{s_1^*, s_2^*, \dots, s_T^*\}$$

$$= \operatorname{argmax}_{s_1, s_2, \dots, s_T} P(s_1, s_2, \dots, s_T | o_1, o_2, \dots, o_T)$$

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$$= \operatorname{argmax}_{s_1, s_2, \dots, s_T} \left[\frac{P(s_1, s_2, \dots, s_T, o_1, o_2, \dots, o_T)}{P(o_1, o_2, \dots, o_T)} \right] \quad \begin{array}{|c|} \hline \text{product} \\ \hline \text{rule} \\ \hline \end{array}$$

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$$\{s_1^*, s_2^*, \dots, s_T^*\}$$

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$$= \operatorname{argmax}_{s_1, s_2, \dots, s_T} P(s_1, s_2, \dots, s_T, o_1, o_2, \dots, o_T) \quad \begin{array}{|c|} \hline \text{no states in} \\ \hline \text{denominator} \\ \hline \end{array}$$

The most likely sequence of hidden states

$$\{s_1^*, s_2^*, \dots, s_T^*\}$$

$$= \operatorname{argmax}_{s_1, s_2, \dots, s_T} P(s_1, s_2, \dots, s_T | o_1, o_2, \dots, o_T)$$

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$$= \operatorname{argmax}_{s_1, s_2, \dots, s_T} \log P(s_1, s_2, \dots, s_T, o_1, o_2, \dots, o_T) \quad \begin{array}{|c|} \hline \text{log is} \\ \hline \text{monotonic} \\ \hline \end{array}$$

The most likely sequence of hidden states

$$\{s_1^*, s_2^*, \dots, s_T^*\}$$

$$= \operatorname{argmax}_{s_1, s_2, \dots, s_T} P(s_1, s_2, \dots, s_T | o_1, o_2, \dots, o_T)$$

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The matrix ℓ^*

- Definition

The matrix ℓ^*

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For a particular sequence of observations $\{o_1, o_2, \dots, o_T\}$, we define the following matrix:

$$\ell_{it}^* = \max_{s_1, s_2, \dots, s_{t-1}} \log P(s_1, s_2, \dots, s_{t-1}, s_t = i, o_1, o_2, \dots, o_t)$$

The matrix ℓ^*

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$$n \text{ rows} \left\{ \begin{bmatrix} \ell_{11}^* & \ell_{12}^* & \cdots & \ell_{1,T-1}^* & \ell_{1T}^* \\ \ell_{21}^* & \ell_{22}^* & \cdots & \ell_{2,T-1}^* & \ell_{2T}^* \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \ell_{n1}^* & \ell_{n2}^* & \cdots & \ell_{n,T-1}^* & \ell_{nT}^* \end{bmatrix} \right.$$

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- Intuition

ℓ_{it}^*	log-probability of the t -step path of hidden states $s_1, s_2, \dots, s_t$ that best explains the observations $o_1, o_2, \dots, o_t$ and ends at state $s_t = i$ at time t
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Computing the matrix ℓ^*

$$\ell_{it}^* = \max_{s_1, s_2, \dots, s_{t-1}} \log P(s_1, s_2, \dots, s_{t-1}, s_t = i, o_1, o_2, \dots, o_t)$$

Computing the matrix ℓ^*

$$\ell_{it}^* = \max_{s_1, s_2, \dots, s_{t-1}} \log P(s_1, s_2, \dots, s_{t-1}, s_t = i, o_1, o_2, \dots, o_t)$$

$$n \text{ rows} \left\{ \begin{bmatrix} \ell_{11}^* \\ \ell_{21}^* \\ \vdots \\ \ell_{n1}^* \end{bmatrix} \quad \ell_{12}^* \quad \cdots \quad \ell_{1,T-1}^* \quad \ell_{1T}^* \right. \\ \left. \ell_{22}^* \quad \cdots \quad \ell_{2,T-1}^* \quad \ell_{2T}^* \right. \\ \vdots \\ \ell_{n2}^* \quad \cdots \quad \ell_{n,T-1}^* \quad \ell_{nT}^* \end{bmatrix}$$

Computing the matrix ℓ^*

$$\ell_{it}^* = \max_{s_1, s_2, \dots, s_{t-1}} \log P(s_1, s_2, \dots, s_{t-1}, s_t = i, o_1, o_2, \dots, o_t)$$

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$$\left. \begin{bmatrix} \ell_{22}^* \\ \vdots \\ \ell_{n2}^* \end{bmatrix} \quad \cdots \quad \ell_{2,T-1}^* \quad \ell_{2T}^* \right.$$
$$\left. \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} \quad \cdots \quad \vdots \quad \vdots \right.$$
$$\left. \begin{bmatrix} \ell_{n,T-1}^* \\ \ell_{nT}^* \end{bmatrix} \right]$$

- First column ($t = 1$)

$$P(s_1 = i, o_1)$$

Computing the matrix ℓ^*

$$\ell_{it}^* = \max_{s_1, s_2, \dots, s_{t-1}} \log P(s_1, s_2, \dots, s_{t-1}, s_t = i, o_1, o_2, \dots, o_t)$$

$$n \text{ rows} \left\{ \begin{bmatrix} \ell_{11}^* \\ \ell_{21}^* \\ \vdots \\ \ell_{n1}^* \end{bmatrix} \quad \ell_{12}^* \quad \cdots \quad \ell_{1,T-1}^* \quad \ell_{1T}^* \right. \\ \left. \begin{bmatrix} \ell_{22}^* \\ \vdots \\ \ell_{n2}^* \end{bmatrix} \quad \cdots \quad \ell_{2,T-1}^* \quad \ell_{2T}^* \right. \\ \left. \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} \quad \cdots \quad \vdots \quad \vdots \right. \\ \left. \begin{bmatrix} \ell_{n,T-1}^* \\ \ell_{nT}^* \end{bmatrix} \right]$$

- First column ($t = 1$)

$$\ell_{i1}^*$$

Computing the matrix ℓ^*

$$\ell_{it}^* = \max_{s_1, s_2, \dots, s_{t-1}} \log P(s_1, s_2, \dots, s_{t-1}, s_t = i, o_1, o_2, \dots, o_t)$$

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$$\left. \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} \quad \cdots \quad \vdots \quad \vdots \right.$$
$$\left. \begin{bmatrix} \ell_{n,T-1}^* \\ \ell_{nT}^* \end{bmatrix} \right]$$

- First column ($t = 1$)

$$\ell_{i1}^* = \log P(S_1 = i, o_1)$$

Computing the matrix ℓ^*

$$\ell_{it}^* = \max_{S_1, S_2, \dots, S_{t-1}} \log P(S_1, S_2, \dots, S_{t-1}, S_t = i, O_1, O_2, \dots, O_t)$$

$$n \text{ rows} \left\{ \begin{bmatrix} \ell_{11}^* \\ \ell_{21}^* \\ \vdots \\ \ell_{n1}^* \end{bmatrix} \quad \ell_{12}^* \quad \cdots \quad \ell_{1,T-1}^* \quad \ell_{1T}^* \right. \\ \left. \begin{bmatrix} \ell_{22}^* \\ \vdots \\ \ell_{n2}^* \end{bmatrix} \quad \cdots \quad \ell_{2,T-1}^* \quad \ell_{2T}^* \right. \\ \left. \begin{bmatrix} \vdots \\ \vdots \\ \vdots \end{bmatrix} \quad \cdots \quad \vdots \quad \vdots \right. \\ \left. \begin{bmatrix} \ell_{n,T-1}^* \\ \ell_{nT}^* \end{bmatrix} \right]$$

- First column ($t = 1$)

$$\begin{aligned} \ell_{i1}^* &= \log P(S_1 = i, O_1) \\ &= \log \left[P(S_1 = i) P(O_1 | S_1 = i) \right] \end{aligned} \quad \boxed{\text{product rule}}$$

Computing the matrix ℓ^*

$$\ell_{it}^* = \max_{S_1, S_2, \dots, S_{t-1}} \log P(S_1, S_2, \dots, S_{t-1}, S_t = i, o_1, o_2, \dots, o_t)$$

$$n \text{ rows} \left\{ \begin{bmatrix} \ell_{11}^* & \ell_{12}^* & \cdots & \ell_{1,T-1}^* & \ell_{1T}^* \\ \ell_{21}^* & \ell_{22}^* & \cdots & \ell_{2,T-1}^* & \ell_{2T}^* \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \ell_{n1}^* & \ell_{n2}^* & \cdots & \ell_{n,T-1}^* & \ell_{nT}^* \end{bmatrix} \right.$$

- First column ($t = 1$)

$$\ell_{i1}^* = \log P(S_1 = i, o_1)$$

$$= \log \left[P(S_1 = i) P(o_1 | S_1 = i) \right]$$

product rule

$$= \log \pi_i + \log b_i(o_1)$$

CPTs

Computing the matrix ℓ^*

- Next columns ($t > 1$)

$$\begin{aligned} \ell_{j,t+1}^* &= \max_{S_1, \dots, S_t} \log P(S_1, \dots, S_t, S_{t+1}=j, O_1, \dots, O_{t+1}) \\ &\stackrel{\text{max}}{=} \max_{S_1, \dots, S_{t-1}} \log P(S_1, \dots, S_{t-1}, \overbrace{S_t=i}^{S_t=i}, \overbrace{O_t=j}^{S_t=j}, O_1, \dots, O_t) \end{aligned}$$

Computing the matrix ℓ^*

- Next columns ($t > 1$)

$$\begin{aligned}\ell_{j,t+1}^* &= \max_{S_1, \dots, S_t} \log P(S_1, \dots, S_t, S_{t+1}=j, O_1, \dots, O_{t+1}) \quad O_{t-1} \\ &= \max_i \max_{S_1, \dots, S_{t-1}} \log P(S_1, \dots, S_{t-1}, S_t=i, S_{t+1}=j, O_1, \dots, O_{t+1})\end{aligned}$$

Handwritten diagram illustrating the decomposition of the probability term in the equation above:

$$P(S_1, \dots, S_{t-1}, S_t=i, O_1, \dots, O_t) \times P(S_{t+1}=j, O_{t+1} | S_t=i)$$

The term $P(S_{t+1}=j, O_{t+1} | S_t=i)$ is further decomposed as:

$$P(S_{t+1}=j | S_t=i) \cdot P(O_{t+1} | S_{t+1}=j)$$

Annotations:

- $P(S_{t+1}=j | S_t=i)$ is labeled "transition (a_{ji})".
- $P(O_{t+1} | S_{t+1}=j)$ is labeled "emission (b_j)".

Computing the matrix ℓ^*

- Next columns ($t > 1$)

$$\begin{aligned}\ell_{j,t+1}^* &= \max_{s_1, \dots, s_t} \log P(s_1, \dots, s_t, S_{t+1}=j, o_1, \dots, o_{t+1}) \\ &= \max_i \max_{s_1, \dots, s_{t-1}} \log P(s_1, \dots, s_{t-1}, S_t=i, S_{t+1}=j, o_1, \dots, o_{t+1}) \\ &= \max_i \max_{s_1, \dots, s_{t-1}} \log \left[P(s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \right.\end{aligned}$$

Computing the matrix ℓ^*

- Next columns ($t > 1$)

$$\begin{aligned}\ell_{j,t+1}^* &= \max_{s_1, \dots, s_t} \log P(s_1, \dots, s_t, S_{t+1}=j, o_1, \dots, o_{t+1}) \\ &= \max_i \max_{s_1, \dots, s_{t-1}} \log P(s_1, \dots, s_{t-1}, S_t=i, S_{t+1}=j, o_1, \dots, o_{t+1}) \\ &= \max_i \max_{s_1, \dots, s_{t-1}} \log \left[P(s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \right] \quad \boxed{\text{product rule}}\end{aligned}$$

Computing the matrix ℓ^*

- Next columns ($t > 1$)

$$\begin{aligned}\ell_{j,t+1}^* &= \max_{s_1, \dots, s_t} \log P(s_1, \dots, s_t, S_{t+1}=j, o_1, \dots, o_{t+1}) \\ &= \max_i \max_{s_1, \dots, s_{t-1}} \log P(s_1, \dots, s_{t-1}, S_t=i, S_{t+1}=j, o_1, \dots, o_{t+1}) \\ &= \max_i \max_{s_1, \dots, s_{t-1}} \log \left[P(s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \boxed{\text{product rule}} \right. \\ &\quad \left. P(S_{t+1}=j | s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \right]\end{aligned}$$

Computing the matrix ℓ^*

- Next columns ($t > 1$)

$$\begin{aligned}\ell_{j,t+1}^* &= \max_{s_1, \dots, s_t} \log P(s_1, \dots, s_t, S_{t+1}=j, o_1, \dots, o_{t+1}) \\ &= \max_i \max_{s_1, \dots, s_{t-1}} \log P(s_1, \dots, s_{t-1}, S_t=i, S_{t+1}=j, o_1, \dots, o_{t+1}) \\ &= \max_i \max_{s_1, \dots, s_{t-1}} \log \left[P(s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \boxed{\text{product rule}} \right. \\ &\quad \left. P(S_{t+1}=j | s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \right. \\ &\quad \left. P(o_{t+1} | s_1, \dots, s_{t-1}, S_t=i, S_{t+1}=j, o_1, \dots, o_t) \right]\end{aligned}$$

Computing the matrix ℓ^*

- Next columns ($t > 1$)

$$\begin{aligned}\ell_{j,t+1}^* &= \max_{s_1, \dots, s_t} \log P(s_1, \dots, s_t, S_{t+1}=j, o_1, \dots, o_{t+1}) \\&= \max_i \max_{s_1, \dots, s_{t-1}} \log P(s_1, \dots, s_{t-1}, S_t=i, S_{t+1}=j, o_1, \dots, o_{t+1}) \\&= \max_i \max_{s_1, \dots, s_{t-1}} \log \left[P(s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \boxed{\text{product rule}} \right. \\&\quad \left. P(S_{t+1}=j | s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \right. \\&\quad \left. P(o_{t+1} | s_1, \dots, s_{t-1}, S_t=i, S_{t+1}=j, o_1, \dots, o_t) \right] \\&= \max_i \max_{s_1, \dots, s_{t-1}} \log \left[P(s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \right.\end{aligned}$$

Computing the matrix ℓ^*

- Next columns ($t > 1$)

$$\begin{aligned}\ell_{j,t+1}^* &= \max_{s_1, \dots, s_t} \log P(s_1, \dots, s_t, S_{t+1}=j, o_1, \dots, o_{t+1}) \\&= \max_i \max_{s_1, \dots, s_{t-1}} \log P(s_1, \dots, s_{t-1}, S_t=i, S_{t+1}=j, o_1, \dots, o_{t+1}) \\&= \max_i \max_{s_1, \dots, s_{t-1}} \log \left[P(s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \boxed{\text{product rule}} \right. \\&\quad \left. P(S_{t+1}=j | s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \right. \\&\quad \left. P(o_{t+1} | s_1, \dots, s_{t-1}, S_t=i, S_{t+1}=j, o_1, \dots, o_t) \right] \\&= \max_i \max_{s_1, \dots, s_{t-1}} \log \left[P(s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \right. \\&\quad \left. P(S_{t+1}=j | S_t=i) \cdot \right]\end{aligned}$$

Computing the matrix ℓ^*

- Next columns ($t > 1$)

$$\begin{aligned}\ell_{j,t+1}^* &= \max_{s_1, \dots, s_t} \log P(s_1, \dots, s_t, S_{t+1}=j, o_1, \dots, o_{t+1}) \\&= \max_i \max_{s_1, \dots, s_{t-1}} \log P(s_1, \dots, s_{t-1}, S_t=i, S_{t+1}=j, o_1, \dots, o_{t+1}) \\&= \max_i \max_{s_1, \dots, s_{t-1}} \log \left[P(s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \boxed{\text{product rule}} \right. \\&\quad \left. P(S_{t+1}=j | s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \right. \\&\quad \left. P(o_{t+1} | s_1, \dots, s_{t-1}, S_t=i, S_{t+1}=j, o_1, \dots, o_t) \right] \\&= \max_i \max_{s_1, \dots, s_{t-1}} \log \left[P(s_1, \dots, s_{t-1}, S_t=i, o_1, \dots, o_t) \cdot \right. \\&\quad \left. P(S_{t+1}=j | S_t=i) \cdot P(o_{t+1} | S_{t+1}=j) \right] \quad \boxed{\text{CI}}\end{aligned}$$

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- Next columns ($t > 1$)

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Computing the matrix ℓ^*

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Computing the matrix ℓ^*

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We have shown how to efficiently compute the matrix ℓ^* one column at a time, from left to right:

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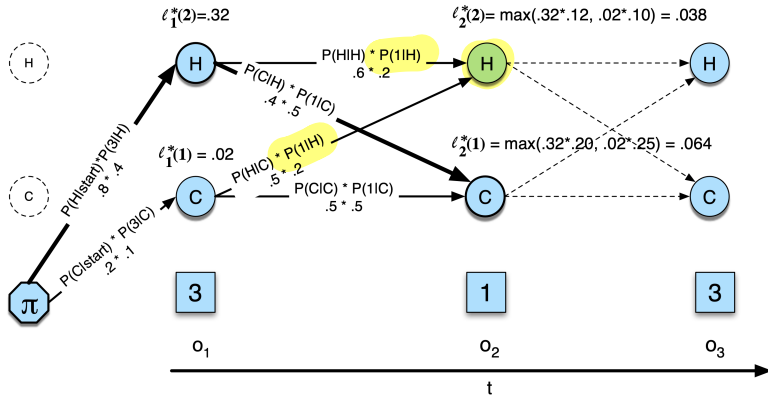
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Example - Viterbi (Fill ℓ^*)



Computing $\{s_1^*, s_2^*, \dots, s_T^*\}$

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- Form one more matrix:

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Intuitively, given the observations $o_1, o_2, \dots, o_{t+1}$, record the most likely state at time t given that $S_{t+1}=j$.

Computing $\{s_1^*, s_2^*, \dots, s_T^*\}$

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Computing $\{s_1^*, s_2^*, \dots, s_T^*\}$

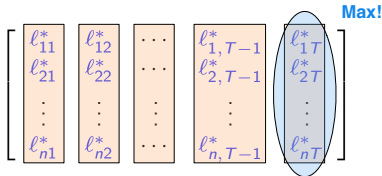
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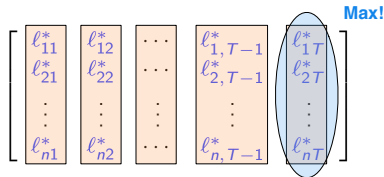
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for $t = T-1$ to 1



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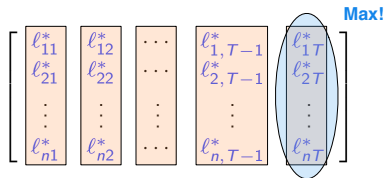
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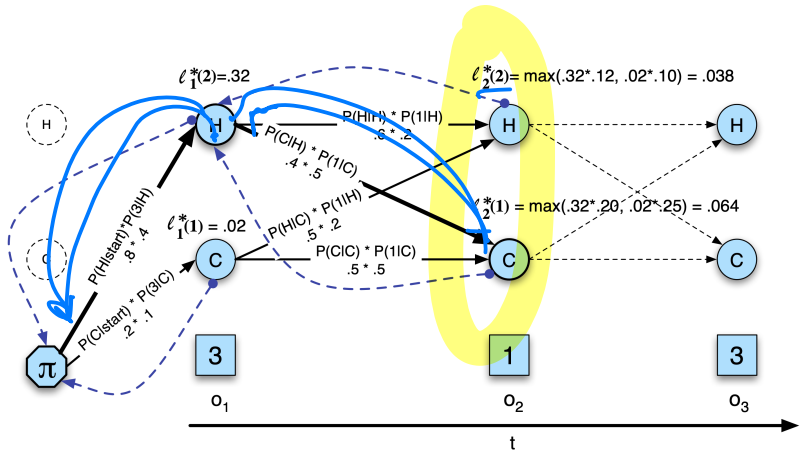
for $t = T-1$ **to** 1

$$s_t^* = \Phi_{t+1}(s_{t+1}^*)$$

end



Example - Viterbi (Backtrack through ℓ^*)



Summary of Viterbi algorithm

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- Fill ℓ^* matrix from left to right:

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$$t = 1$$

Summary of Viterbi algorithm

- Fill ℓ^* matrix from left to right:

$$\boxed{t = 1} \quad \ell_{i1}^* = \log \pi_i + \log b_i(o_1)$$

Summary of Viterbi algorithm

- Fill ℓ^* matrix from left to right:

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$$\boxed{t > 1}$$

Summary of Viterbi algorithm

- Fill ℓ^* matrix from left to right:

$$\boxed{t = 1} \quad \ell_{i1}^* = \log \pi_i + \log b_i(o_1)$$

$$\boxed{t > 1} \quad \ell_{j,t+1}^* = \max_i \left[\ell_{it}^* + \log a_{ij} \right] + \log b_j(o_{t+1})$$

Summary of Viterbi algorithm

- Fill ℓ^* matrix from left to right:

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- Backtrack through ℓ^* from right to left:

Summary of Viterbi algorithm

- Fill ℓ^* matrix from left to right:

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$$\boxed{t = T}$$

Summary of Viterbi algorithm

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- Backtrack through ℓ^* from right to left:

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- Fill ℓ^* matrix from left to right:

$$\boxed{t = 1} \quad \ell_{i1}^* = \log \pi_i + \log b_i(o_1)$$

$$\boxed{t > 1} \quad \ell_{j,t+1}^* = \max_i \left[\ell_{it}^* + \log a_{ij} \right] + \log b_j(o_{t+1})$$

- Backtrack through ℓ^* from right to left:

$$\boxed{t = T} \quad s_T^* = \operatorname{argmax}_i \left[\ell_{iT}^* \right]$$

$$\boxed{t < T}$$

Summary of Viterbi algorithm

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This algorithm is an instance of **dynamic programming**.

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This algorithm is an instance of **dynamic programming**.



Sometimes $\{s_1^*, s_2^*, \dots, s_T^*\}$ is called the **Viterbi path**.

Which of the following statements is true(in terms of asymptotic time complexity)?

- A. The Viterbi algorithm generally runs faster than the Forward algorithm because it finds a single most likely sequence.
- B. The Forward algorithm generally runs faster than the Viterbi algorithm because it computes probabilities for all possible sequences.
-  C. Both algorithms have similar running times.
- D. The Viterbi algorithm runs slower than the Forward algorithm due to its additional backtracking step.

What is the running time for Viterbi algorithm?

A. $O(n)$

B. $O(n^2)$

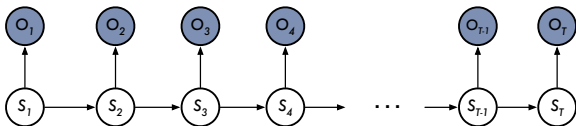
C. $O(Tn^2)$

D. $O(T^2n^4)$

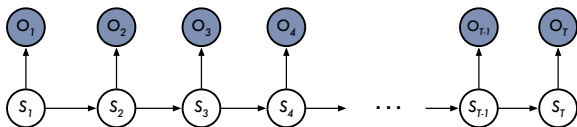
E. $O(n^T)$

Learning in HMMs

Learning in HMMs

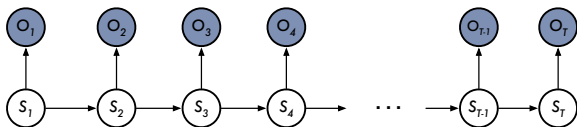


Learning in HMMs



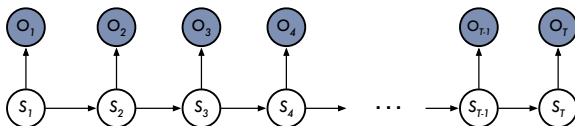
Given: one or more sequences of observations $\{o_1, o_2, \dots, o_T\}$.

Learning in HMMs



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For simplicity, we'll assume just one.

Learning in HMMs

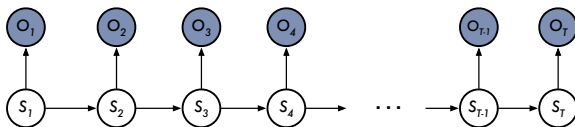


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For simplicity, we'll assume just one.

Goal: estimate $\{\pi_i, a_{ij}, b_{ik}\}$ to maximize $P(o_1, o_2, \dots, o_T)$,

Learning in HMMs

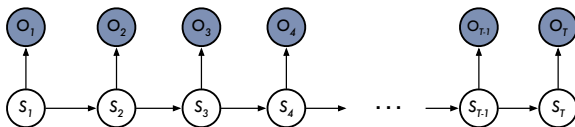


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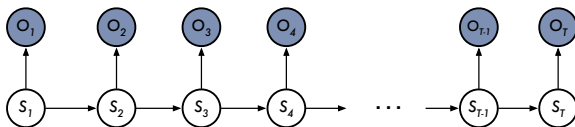
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Assume: the cardinality n of the hidden state space is fixed.

Learning in HMMs



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Assume: the cardinality n of the hidden state space is fixed.

$$s_t \in \{1, 2, \dots, n\}$$

How do we estimate $\{\pi_i, a_{ij}, b_{ik}\}$?

EM Algorithm!

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EM Algorithm!

How EM works in general:

To re-estimate $P(X_i = x | \text{pa}_i = \pi)$ in the M-step,
we must compute $P(X_i = x, \text{pa}_i = \pi | V)$ in the E-step.

EM algorithm for HMMs

- CPTs to re-estimate:

EM algorithm for HMMs

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$$\pi_i = P(S_1=i)$$

EM algorithm for HMMs

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EM algorithm for HMMs

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$$P(S_1=i|o_1, o_2, \dots, o_T)$$

EM algorithm for HMMs

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$$P(O_t=k, S_t=i|o_1, o_2, \dots, o_T) = \underbrace{l(o_t, k)} P(S_t=i|o_1, o_2, \dots, o_T)$$

EM algorithm for HMMs

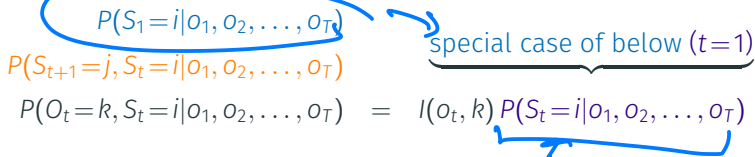
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$$P(S_1=i|o_1, o_2, \dots, o_T)$$

special case of below ($t=1$)

$$P(S_{t+1}=j, S_t=i|o_1, o_2, \dots, o_T)$$
$$P(O_t=k, S_t=i|o_1, o_2, \dots, o_T) = l(o_t, k) P(S_t=i|o_1, o_2, \dots, o_T)$$

EM algorithm for HMMs

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$$a_{ij} = P(S_{t+1}=j|S_t=i)$$

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- E-step in HMMs must compute:

$$\begin{aligned} &P(S_1=i|o_1, o_2, \dots, o_T) \\ &P(S_{t+1}=j, S_t=i|o_1, o_2, \dots, o_T) \quad \underbrace{\text{special case of below } (t=1)} \\ &P(O_t=k, S_t=i|o_1, o_2, \dots, o_T) = l(o_t, k) P(S_t=i|o_1, o_2, \dots, o_T) \end{aligned}$$

How to efficiently compute these posteriors?

Computing $P(S_t = i | o_1, \dots, o_T)$

Computing $P(S_t=i|o_1,\dots,o_T)$

$$P(S_t=i|o_1,\dots,o_T) =$$

Computing $P(S_t=i|o_1,\dots,o_T)$

$$P(S_t=i|o_1,\dots,o_T) = \frac{P(S_t=i, o_1, o_2, \dots, o_T)}{P(o_1, o_2, \dots, o_T)}$$

product rule

Computing $P(S_t=i|o_1,\dots,o_T)$

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- Numerator

$$\begin{aligned} P(S_t=i, o_1, o_2, \dots, o_T) \\ = P(o_1, \dots, o_t, S_t=i) P(o_{t+1}, \dots, o_T | S_t=i, o_1, \dots, o_t) \quad \text{product rule} \end{aligned}$$

Computing $P(S_t=i|o_1,\dots,o_T)$

$$P(S_t=i|o_1,\dots,o_T) = \frac{P(S_t=i, o_1, o_2, \dots, o_T)}{P(o_1, o_2, \dots, o_T)}$$

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$$\begin{aligned} &P(S_t=i, o_1, o_2, \dots, o_T) \\ &= P(o_1, \dots, o_t, S_t=i) P(\underline{o_{t+1}, \dots, o_T} | S_t=i, \cancel{o_1, \dots, o_t}) \\ &= P(o_1, \dots, o_t, S_t=i) P(o_{t+1}, \dots, o_T | S_t=i) \end{aligned}$$

product rule

conditional independence

Computing $P(S_t=i|o_1,\dots,o_T)$

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product rule

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$$\begin{aligned} &P(S_t=i, o_1, o_2, \dots, o_T) \\ &= P(o_1, \dots, o_t, S_t=i) P(o_{t+1}, \dots, o_T | S_t=i, o_1, \dots, o_t) \quad \text{product rule} \\ &= P(o_1, \dots, o_t, S_t=i) P(o_{t+1}, \dots, o_T | S_t=i) \quad \text{conditional independence} \\ &= \alpha_{it} P(o_{t+1}, \dots, o_T | S_t=i) \end{aligned}$$

Backward Algorithm

We need one more matrix ...

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Analogous to $\alpha_{it} = P(o_1, o_2, \dots, o_t, S_t = i),$

We need one more matrix ...

Analogous to $\alpha_{it} = P(o_1, o_2, \dots, o_t, S_t = i),$

define $\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i).$

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define $\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$.

$$n \text{ rows} \left\{ \begin{bmatrix} \beta_{11} & \beta_{12} & \cdots & \beta_{1,T-1} & \beta_{1T} \\ \beta_{21} & \beta_{22} & \cdots & \beta_{2,T-1} & \beta_{2T} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \beta_{n1} & \beta_{n2} & \cdots & \beta_{n,T-1} & \beta_{nT} \end{bmatrix} \right.$$

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Understand the **differences** between these matrices:

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define $\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i).$

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Understand the **differences** between these matrices:

- α_{it} predicts observations up to and including time t .
- β_{it} predicts observations from **time $t + 1$** to time T .

Computing $\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$

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- Last column ($t = T$)

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- Last column ($t = T$)

$$\beta_{iT} = P(\text{---} | \underline{S_T = i})$$

What does this mean?

Computing $\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$

- Last column ($t = T$)

$$\beta_{iT} = P(\text{---} | S_T = i) \quad \text{What does this mean?}$$

Note: β_{it} computes the probability of the future given $S_t = i$.

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But we don't see *any* observations beyond time T .

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Put another way, the future after time T is unspecified.

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What is the probability of some unspecified future occurring?

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Note: β_{it} computes the probability of the future given $S_t = i$.

But we don't see *any* observations beyond time T .
Put another way, the future after time T is unspecified.

What is the probability of some unspecified future occurring?

By definition, we set:

$$\beta_{iT} = 1 \quad \text{for all } i \in \{1, 2, \dots, n\}$$

Computing $\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$

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- Previous columns ($t < T$)

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$$\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$$

=

Computing $\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$

- Previous columns ($t < T$)

$$\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$$

$$= \sum_{j=1}^n P(S_{t+1} = j, o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$$

marginalization

=

Computing $\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$

- Previous columns ($t < T$)

$$\begin{aligned}\beta_{it} &= P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i) \\ &= \sum_{j=1}^n P(S_{t+1} = j, o_{t+1}, o_{t+2}, \dots, o_T | S_t = i) \\ &= \sum_{j=1}^n \left[P(S_{t+1} = j | S_t = i) \cdot \right.\end{aligned}$$

marginalization

Computing $\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$

- Previous columns ($t < T$)

$$\begin{aligned}\beta_{it} &= P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i) \\ &= \sum_{j=1}^n P(S_{t+1} = j, o_{t+1}, o_{t+2}, \dots, o_T | S_t = i) \\ &= \sum_{j=1}^n \left[P(S_{t+1} = j | S_t = i) \cdot \right. \\ &\quad \left. P(o_{t+1} | S_t = i, S_{t+1} = j) \cdot \right.\end{aligned}$$

marginalization

Computing $\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$

- Previous columns ($t < T$)

$$\begin{aligned}\beta_{it} &= P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i) \\&= \sum_{j=1}^n P(S_{t+1} = j, o_{t+1}, o_{t+2}, \dots, o_T | S_t = i) \quad \boxed{\text{marginalization}} \\&= \sum_{j=1}^n \left[P(S_{t+1} = j | S_t = i) \cdot \right. \\&\quad \left. P(o_{t+1} | S_t = i, S_{t+1} = j) \cdot \right. \\&\quad \left. P(o_{t+2}, \dots, o_T | S_t = i, S_{t+1} = j, o_{t+1}) \right] \quad \boxed{\text{product rule}} \\&= \end{aligned}$$

Computing $\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$

- Previous columns ($t < T$)

$$\begin{aligned}\beta_{it} &= P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i) \\&= \sum_{j=1}^n P(S_{t+1} = j, o_{t+1}, o_{t+2}, \dots, o_T | S_t = i) \quad \boxed{\text{marginalization}} \\&= \sum_{j=1}^n \left[P(S_{t+1} = j | S_t = i) \cdot \right. \\&\quad \left. P(o_{t+1} | S_t = i, S_{t+1} = j) \cdot \right. \\&\quad \left. P(o_{t+2}, \dots, o_T | S_t = i, S_{t+1} = j, o_{t+1}) \right] \quad \boxed{\text{product rule}} \\&= \sum_{j=1}^n \left[P(S_{t+1} = j | S_t = i) \right.\end{aligned}$$

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Computing $\beta_{it} = P(o_{t+1}, o_{t+2}, \dots, o_T | S_t = i)$

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Backward algorithm

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The backward algorithm fills in the matrix of β_{it} elements one column at a time:

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$$n \text{ rows} \left\{ \begin{bmatrix} \beta_{11} & \beta_{12} & \cdots & \beta_{1,T-1} & 1 \\ \beta_{21} & \beta_{22} & \cdots & \beta_{2,T-1} & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \beta_{n1} & \beta_{n2} & \cdots & \beta_{n,T-1} & 1 \end{bmatrix} \right.$$

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Computing $P(S_t = i | o_1, \dots, o_T)$

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$$P(S_t=i|o_1, \dots, o_T) = \frac{P(S_t=i, o_1, o_2, \dots, o_T)}{P(o_1, o_2, \dots, o_T)} \quad \text{product rule}$$

- Numerator

$$\begin{aligned} &P(S_t=i, o_1, o_2, \dots, o_T) \\ &= P(o_1, \dots, o_t, S_t=i) P(o_{t+1}, \dots, o_T | S_t=i, o_1, \dots, o_t) \quad \text{product rule} \\ &= P(o_1, \dots, o_t, S_t=i) P(o_{t+1}, \dots, o_T | S_t=i) \quad \text{conditional independence} \\ &= \alpha_{it} P(o_{t+1}, \dots, o_T | S_t=i) \end{aligned}$$

Computing $P(S_t=i|o_1, \dots, o_T)$

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Next lecture: EM with the forward-backward algorithm.

That's all folks!